

Compression and Estimation: Transformers and Canonical Problems in Information Theory

Krishna Narayanan

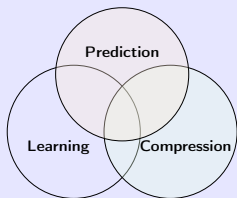
Co-Director, Texas A&M Initiative for Connected Intelligence

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Perspectives and Trends

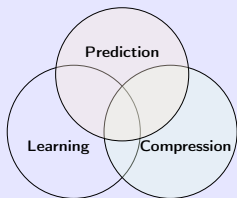
The Triad



What do advances in learning mean for information compression/communication/hiding?

Perspectives and Trends

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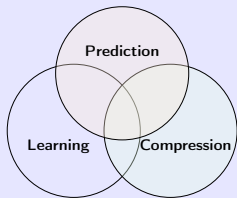
AI-generated Content



- ▶ AI-generated content continues to increase
- ▶ Newer paradigms for compression

Perspectives and Trends

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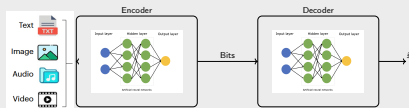
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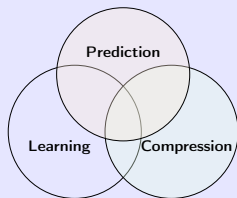
End-to-End Learning is not only research



- ▶ JPEG AI is an international standard
- ▶ JPEG XE for lossless coding of event-driven data

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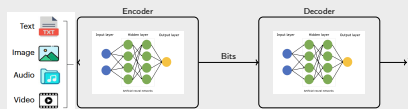
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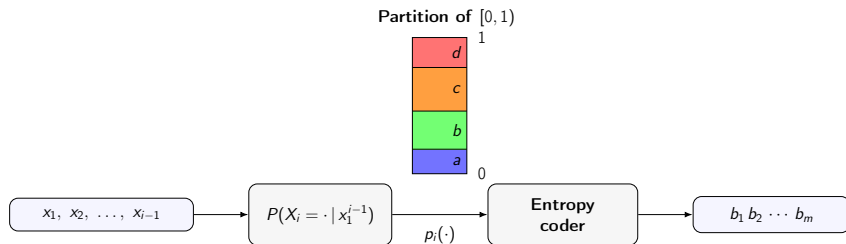
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AI Native Wireless / 6G

- ▶ Channel estimation based on AI
- ▶ Channel state information compression
- ▶ Joint source channel has been revitalized
- ▶ Learned neural receivers

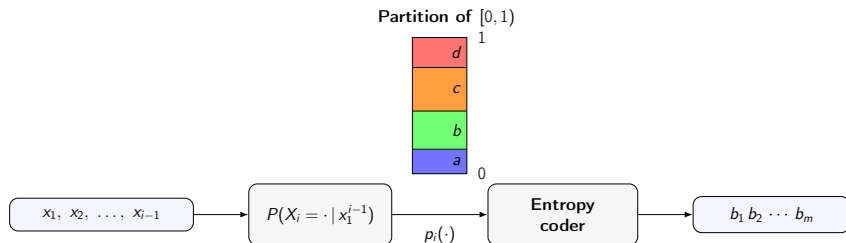


Lossless Compression Setting



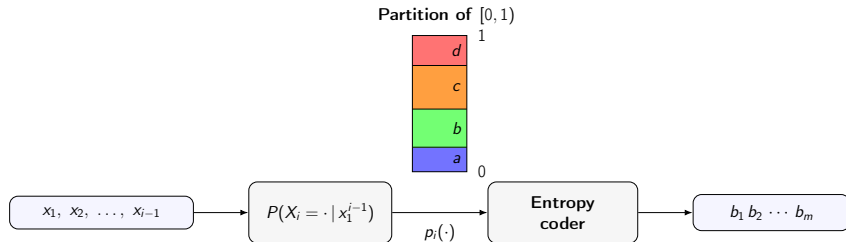
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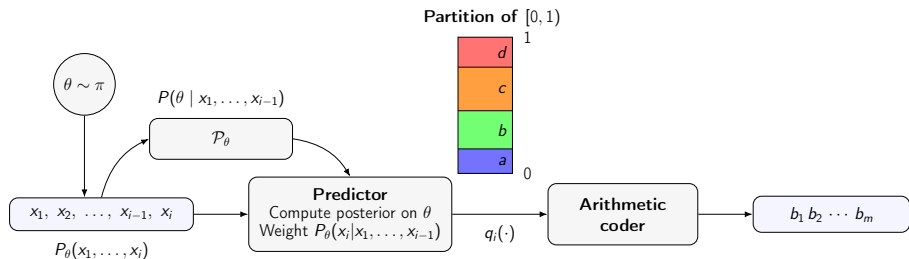
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 - Factorize $P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | x_1, x_2, \dots, x_{i-1})$
 - Combine with a good entropy coder

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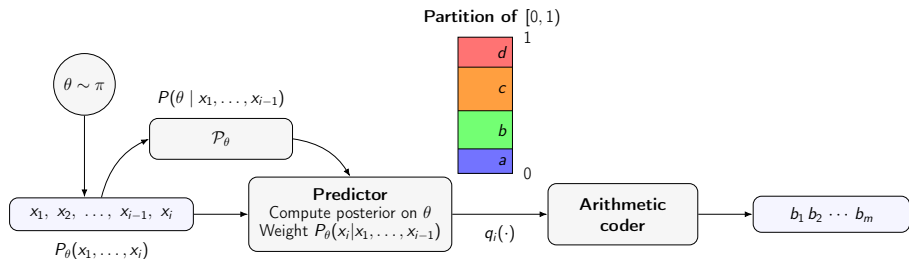
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- ▶ $P(x_1, x_2, \dots, x_n)$ is not known - universal setting
 - Long line of work in IT
 - Treated extensively in Merhav and Feder, Universal Prediction, IT, 1998

Prior distribution π is known



- For a given θ , Redundancy: $\frac{1}{n} \{ \mathbb{E}[-\log q(X_1^n)] + \mathbb{E}[\log P_\theta(X_1^n)] \} = \frac{1}{n} D_n(P_\theta \| q)$

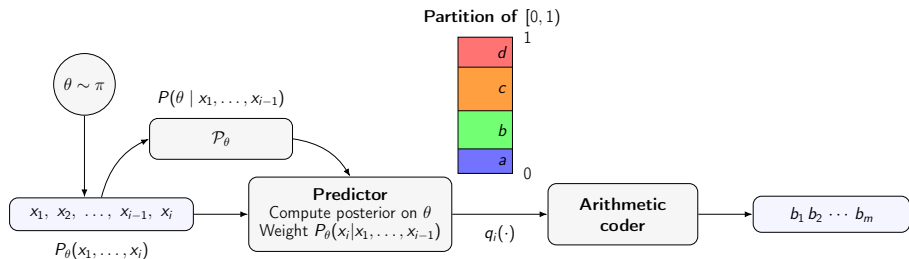
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$$q_i(x_i | \underbrace{x_1, \dots, x_{i-1}}_{\text{context/prompt}}) = \int P(\theta | x_1, \dots, x_{i-1}) P_\theta(x_i | x_1, \dots, x_{i-1}) d\theta$$

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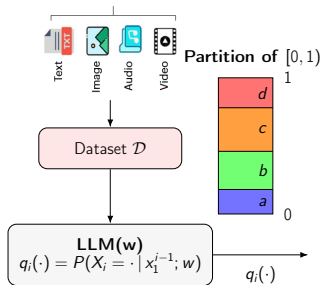


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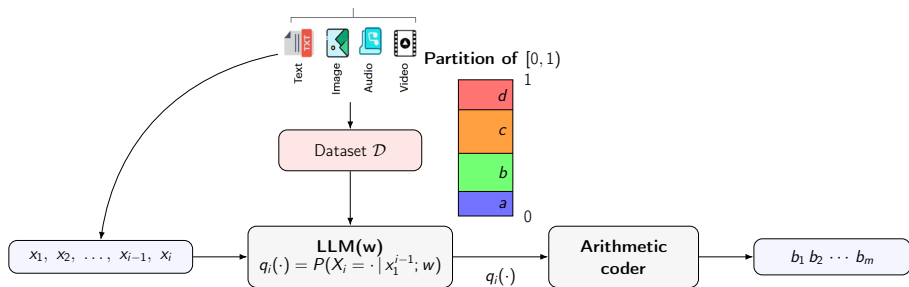
- ▶ *Computationally tractable only in some cases*
- ▶ Replace $\int D_n(P_\theta \| q) \pi(\theta) d\theta$ by $-\frac{1}{N} \sum_{i=1}^N \log q_i(x_i | x_1, \dots, x_{i-1}; w)$ over $\mathcal{D}$

Setting 3 - Data Driven Lossless Compression



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- ▶ Models may memorize/compress the training data
- ▶ During testing, it is assumed that a *new* realization is generated

Shannon's Prediction Game (1951)²

"I tried to explain mutual information at a party once.

Nobody was impressed.

But given the context, it was totally - - - - -"

- Humans to guess the next token and compress the ranks

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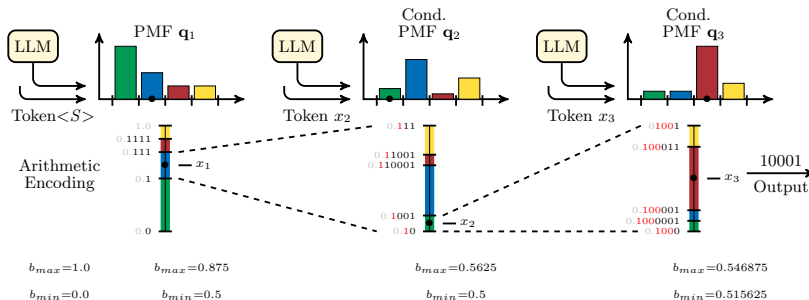
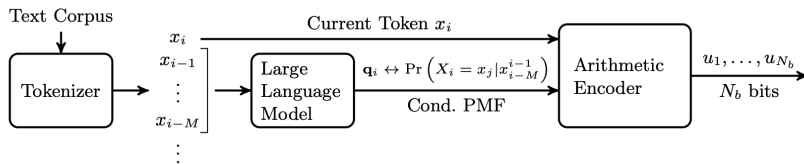
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- ▶ *We thought why not use an LLM for prediction?*
- ▶ Indeed, recently neural networks have been used for prediction¹

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LLMZip - Encoder

Example: Alphabet $\{G, B, R, Y\}$ and $\underline{s} = (B, G, R)$



LLMZip Compression Performance⁷

Table: Compression performance (bits per char) of LLaMA models, other SOTA compressors, and known bounds.

Dataset	LLaMa3 - 70B + AC	LLaMA2 - 7B + AC	ZPAQ	paq8	zlib	Lower / Upper bounds ³	Upper bound ⁴	Upper bound ⁵
text8		0.693	1.52	1.3	—	0.6 / 1.4	1.2	1.05
Gutenberg			—	—	2.55	0.6 / 1.3	1.2	—
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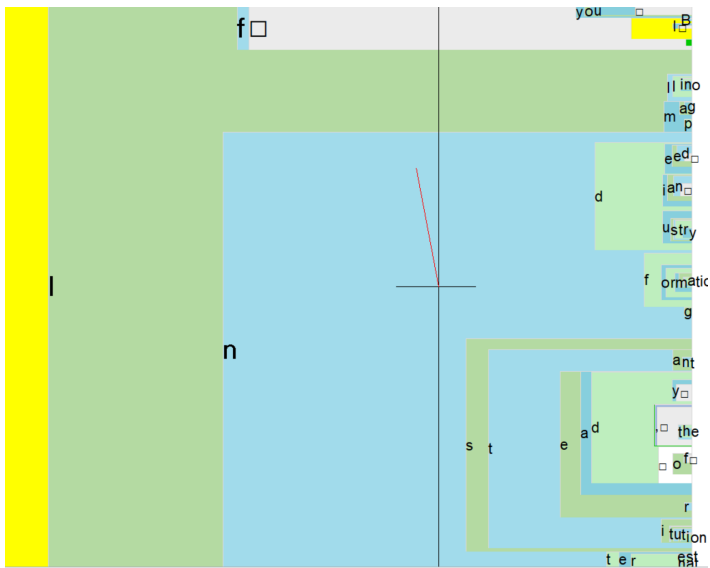
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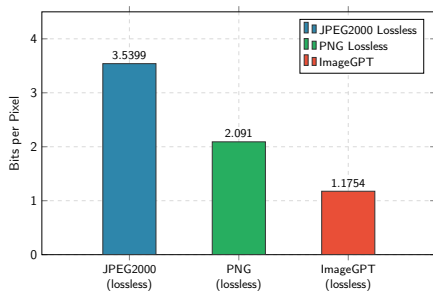
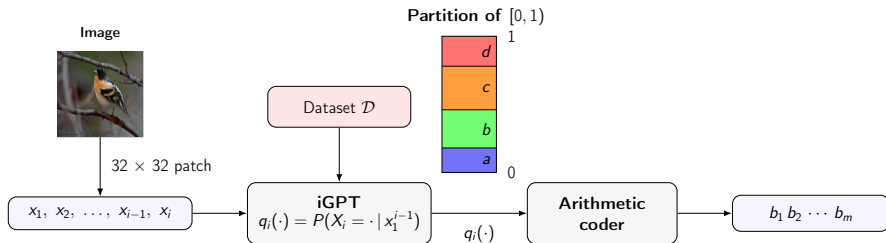
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Dasher



Lossless Image Compression



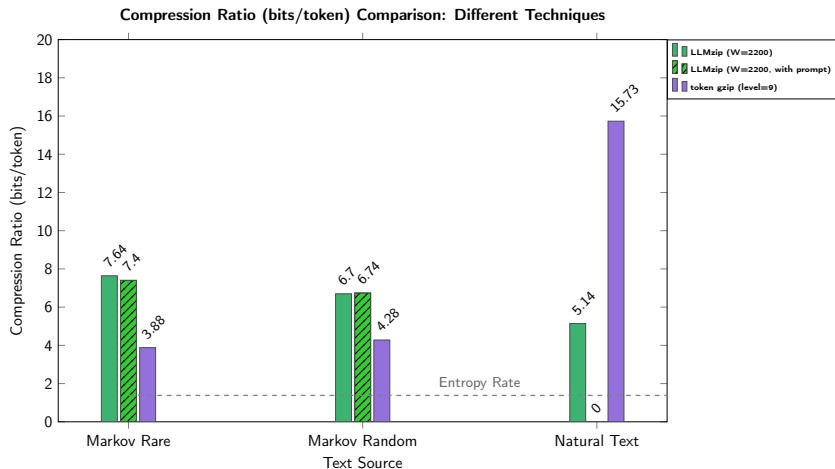
Is there Out of Distribution Learning?⁸

Chunk	Compressor	Raw Compression Rate (%)				Adjusted Compression Rate (%)			
		enwik9	ImageNet	LibriSpeech	Random	enwik9	ImageNet	LibriSpeech	Random
∞	gzip	32.3	70.7	36.4	100.0	32.3	70.7	36.4	100.0
	LZMA2	23.0	57.9	29.9	100.0	23.0	57.9	29.9	100.0
	PNG	42.9	58.5	32.2	100.0	42.9	58.5	32.2	100.0
	FLAC	89.5	61.9	30.9	107.8	89.5	61.9	30.9	107.8
2048	gzip	48.1	68.6	38.5	100.1	48.1	68.6	38.5	100.1
	LZMA2	50.0	62.4	38.2	100.0	50.0	62.4	38.2	100.0
	PNG	80.6	61.7	37.6	103.2	80.6	61.7	37.6	103.2
	FLAC	88.9	60.9	30.3	107.2	88.9	60.9	30.3	107.2
	Transformer 200K	30.9	194.0	146.6	195.5	30.9	194.0	146.6	195.5
	Transformer 800K	21.7	185.1	131.1	200.1	21.9	185.3	131.3	200.3
	Transformer 3.2M	17.0	215.8	228.2	224.0	17.7	216.5	228.9	224.7
	Llama 2 (7B)	8.9	53.4	23.1	103.2	1408.9	1453.4	1423.1	1503.2
	Chinchilla 1B	11.3	62.2	24.9	108.8	211.3	262.2	224.9	308.8
	Chinchilla 7B	10.2	54.7	23.6	101.6	1410.2	1454.7	1423.6	1501.6
	Chinchilla 70B	8.3	48.0	21.0	100.8	14008.3	14048.0	14021.0	14100.8

(see Table 1). In contrast, general-purpose compressors, such as gzip, offer good performance on a wide range of data sources. Surprisingly, **large language models, while trained primarily on text, also appear to be general-purpose compressors**, as they outperform all other compressors, even on image and audio data (see Table 1). Note that **these models have not been trained on this kind of data**: for Chinchilla, Appendix A. of Hoffmann et al. (2022) states that the training dataset consists of a mix of internet text data (Wikipedia, websites, github) and books. However, **it is still possible (but unlikely) that some images or audio samples were encoded into text on some websites**. Thus, pretrained models achieve their impressive compression performance by conditioning a (meta-)trained model to a particular task at hand via in-context learning (Genewein et al., 2023). In contrast, smaller Transformers, trained manually on enwik8, only achieve good compression rates on similar Wikipedia

Out of Distribution Experiment

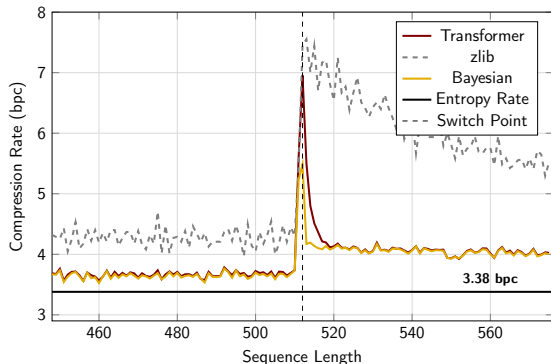
We picked 10 tokens with low probabilities and constructed a Markov chain to create an OOD sequence. There appears to be no learning by the LLM.



Transformers are near optimal Bayesian Estimators

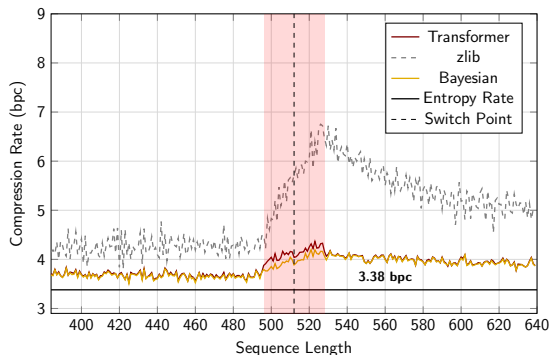
- ▶ Several small examples - Laplace's +1 rule, mixture of Markov sources
- ▶ Transformers can learn context-trees⁹
- ▶ Time-varying source that switches from Markov1 to Markov2

Compression Rate vs. Sequence Length



(a) Switch Point Window = 1

Compression Rate vs. Sequence Length

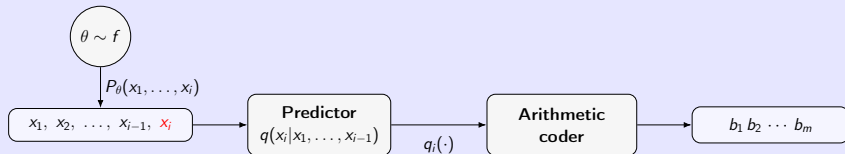


(b) Switch Point Window = 32

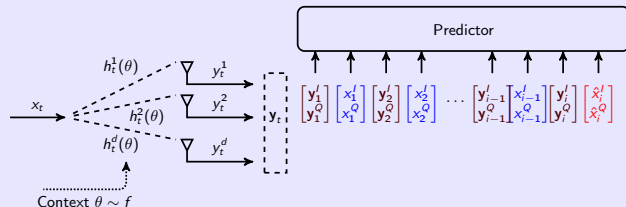
⁹R. Zhou, C. Tian, and S. Diggavi (n.d.). "Transformers Learn to Compress Variable-order Markov Chains in-Context". In: *Workshop on Machine Learning and Compression, NeurIPS 2024*

Estimation as the Central Problem

Lossless Compression



Symbol Estimation in Wireless Communications



- ▶ $\mathbf{y} = Q(h\mathbf{x} + \mathbf{n})$ - h comes from a complicated prior
- ▶ Pilots - $X_1, \dots, X_{i-1} \rightarrow$ Received signals $\mathbf{y}_1, \dots, \mathbf{y}_{i-1}$
- ▶ Given $(\mathbf{y}_1, x_1), (\mathbf{y}_2, x_2), \dots, (\mathbf{y}_i, ?)$

Symbol Estimation in Wireless Communications

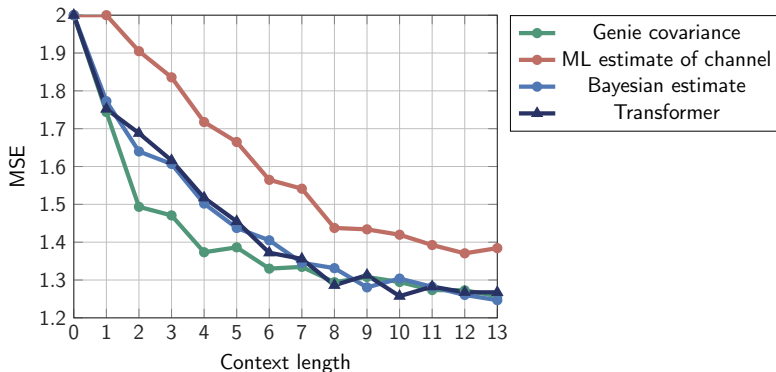
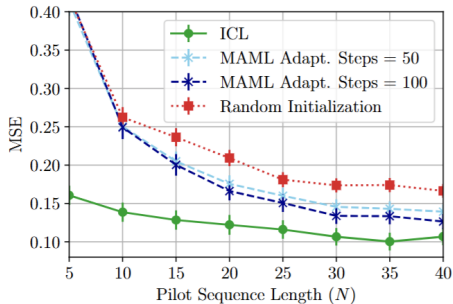
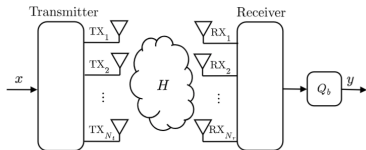


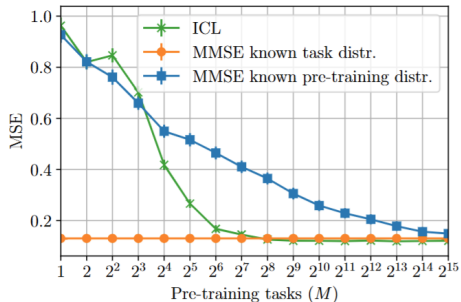
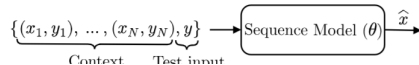
Figure: $y = hx + n$, h - Mixture Gaussian; Given $(y_1, x_1), (y_2, x_2), \dots, (y_n, ?)$

ICL for Quantized MIMO

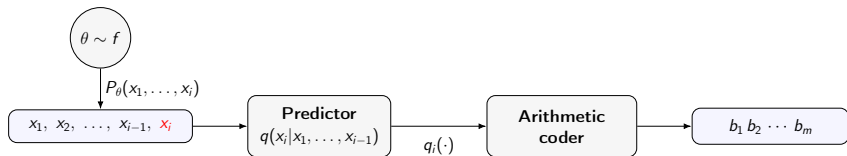


② Online: ICL-Based Equalization

- Receive context $(x_1, y_1), \dots, (x_N, y_N)$ and test input y
- Predict output $\hat{x}$ corresponding to the test input y

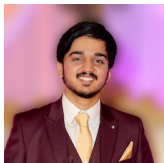


Concluding Remarks



- ▶ Architectural principle from IT
- ▶ The loss function to minimize and the function to approximate from IT
- ▶ Bypassing the modeling came from a data-driven approach
- ▶ Data, Compute, Training tricks from ML community
- ▶ *LLMs approximate the Bayes predictor well in data rich regions*
- ▶ *I don't see evidence of meta learning or OOD learning*

Collaborators



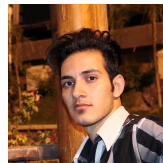
Kaushik
Valmeekam



Vishnu
Kunde



Yu-Shin
Huang



Mahdi
Farahbakhsh



Jean-Francois
Chamberland



Dileep
Kalathil



Chao
Tian

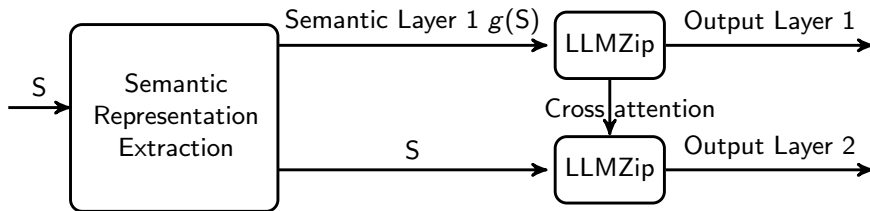


Srinivas
Shakkottai

Compression with Summaries as Side Information

- ▶ LLMs are good at summarizing text
- ▶ For a text S , let $g(S)$ denote a summary
- ▶ Consider 2-step compression: compress $g(S)$ then compress $S|g(S)$

$$H(S, g(S)) = H(S) + H(g(S)|S) = H(g(S)) + H(S|g(S))$$



- ▶ Summarization provides a natural way to perform semantic compression

Preliminary Results

Table: Compression performance (bpc) of LLaMa2 +AC with summary and $M = 511$ on a recently published paper in arXiv

<i>Type</i>	Summary		Text		LLaMa2+AC	LLaMa2+AC
	Toks	Chars	Toks	Chars	SI @ Enc & Dec	SI @ Enc
No Summary	0	0	12578	43787	0.6019	0.6019
Abstract	252	1312	12578	43787	0.5846	0.5997
Summary 1	219	1095	12578	43787	0.5849	0.5989
Summary 2	473	2233	12578	43787	0.5672	0.5965
Summary 3	428	1918	12578	43787	0.5566	0.5880
Summary 4	945	4919	12578	43787	0.5556	0.6064
Paragraph 1	281	1255	12578	43787	0.5981	0.6194

Lossy Compression of Audio¹¹

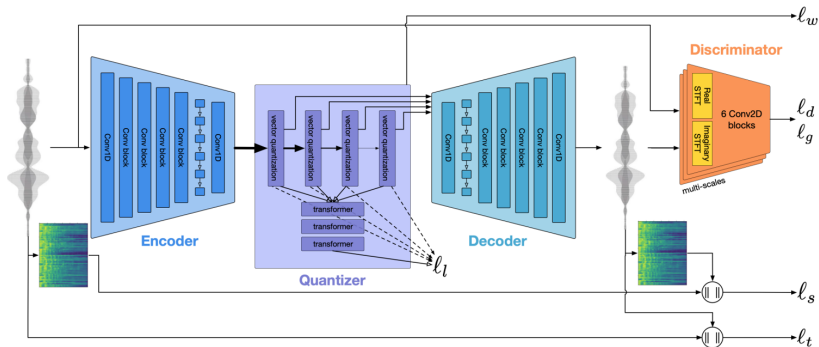


Figure 1: ENCODEC : an encoder decoder codec architecture which is trained with reconstruction (ℓ_f and ℓ_t) as well as adversarial losses (ℓ_g for the generator and ℓ_d for the discriminator). The residual vector quantization commitment loss (ℓ_w) applies only to the encoder. Optionally, we train a small Transformer language model for entropy coding over the quantized units with ℓ_t , which reduces bandwidth even further.

¹¹A. Défossez et al. (2022). *High Fidelity Neural Audio Compression*. [arXiv: 2210.13438 \[eess.AS\]](https://arxiv.org/abs/2210.13438).

Lossy Compression of Audio Results¹²

Table 1: MUSHRA scores for Opus, EVS, Lyra-v2, and ENCODEC for various bandwidths under the streamable setting. Results are reported across different audio categories (clean speech, noisy speech, and music), sampled at 24 kHz. We report mean scores and 95% confidence intervals. For ENCODEC, we also report the average bandwidth after using the entropy coding described in Section 3.3.

Model	Bandwidth	Entropy Coded	Clean Speech	Noisy Speech	Music Set-1	Music Set-2
Reference	-	-	95.5 $\pm$ 1.6	93.9 $\pm$ 1.8	93.2 $\pm$ 2.5	97.1 $\pm$ 1.3
Opus	6.0 kbps	-	30.1 $\pm$ 2.8	19.1 $\pm$ 5.9	20.6 $\pm$ 5.8	17.9 $\pm$ 5.3
Opus	12.0 kbps	-	76.5 $\pm$ 2.3	61.9 $\pm$ 2.1	77.8 $\pm$ 3.2	65.4 $\pm$ 2.7
EVS	9.6 kbps	-	84.4 $\pm$ 2.5	80.0 $\pm$ 2.4	89.9 $\pm$ 2.3	87.7 $\pm$ 2.3
Lyra-v2	3.0 kbps	-	53.1 $\pm$ 1.9	52.0 $\pm$ 4.7	69.3 $\pm$ 3.3	42.3 $\pm$ 3.5
Lyra-v2	6.0 kbps	-	66.2 $\pm$ 2.9	59.9 $\pm$ 3.3	75.7 $\pm$ 2.6	48.6 $\pm$ 2.1
ENCODEC	1.5 kbps	0.9 kbps	49.2 $\pm$ 2.4	41.3 $\pm$ 3.6	68.2 $\pm$ 2.2	66.5 $\pm$ 2.3
ENCODEC	3.0 kbps	1.9 kbps	67.0 $\pm$ 1.5	62.5 $\pm$ 2.3	89.6 $\pm$ 3.1	87.8 $\pm$ 2.9
ENCODEC	6.0 kbps	4.1 kbps	83.1 $\pm$ 2.7	69.4 $\pm$ 2.3	92.9 $\pm$ 1.8	91.3 $\pm$ 2.1
ENCODEC	12.0 kbps	8.9 kbps	90.6 $\pm$ 2.6	80.1 $\pm$ 2.5	91.8 $\pm$ 2.5	92.9 $\pm$ 1.2

¹²A. Défossez et al. (2022). *High Fidelity Neural Audio Compression*. [arXiv: 2210.13438 \[eess.AS\]](https://arxiv.org/abs/2210.13438).